

NS5

The solid cylinder of radius a shown in the figure moves vertically with velocity U , dragging in its motion a coaxial cylindrical layer of liquid of constant uniform thickness h . The liquid has constant values of its density ρ , viscosity μ , thermal conductivity k , and specific heat c . The air surrounding the liquid has a density $\rho_a \ll \rho$ and viscosity $\mu_a \ll \mu$, so that in the first approximation it moves without significant pressure differences, exerting on the liquid a negligible viscous force. Because of the symmetry of the problem, $\partial/\partial\theta = 0$ and the only nonzero velocity component is v_z .

1. Obtain the velocity $v_z(r, z)$ and pressure $p(r, z)$ in the liquid.
2. Compute the force on the cylinder (per unit length).
3. Calculate the volume flux Q dragged by the cylinder, as well as the value of U for which $Q = 0$.
4. Assuming that the thermal conductivity of air is very small compared with that of the liquid and that the cylinder is at constant uniform temperature T_c , obtain the temperature distribution in the liquid $T(r)$ as well as the conductive heat flux to the wall.

1) $\frac{\partial v_z}{\partial z} = 0 \rightarrow v_z = v_z(r)$
 r -mom: $0 = -\frac{\partial p}{\partial r} \Rightarrow p = p(z) = p_a$ $r = a+h, p = p_a$

$0 = \mu \frac{1}{r} \frac{d}{dr} \left(r \frac{dv_z}{dr} \right) - \rho g, \quad r=a: v_z = U$
 $r=a+h: \frac{dv_z}{dr} = 0$

$v_z = U + \frac{g}{2\nu} \left[\frac{r^2 - a^2}{2} - (a+h)^2 \ln\left(\frac{r}{a}\right) \right]$ $g \downarrow$

2) $\oint \vec{\tau} \cdot \vec{n} d\sigma = \mu \frac{dv_z}{dr} \bigg|_{r=a} 2\pi a \vec{e}_z = -2\pi a \rho g h \left[1 + \frac{1}{2a} \right] \vec{e}_z$

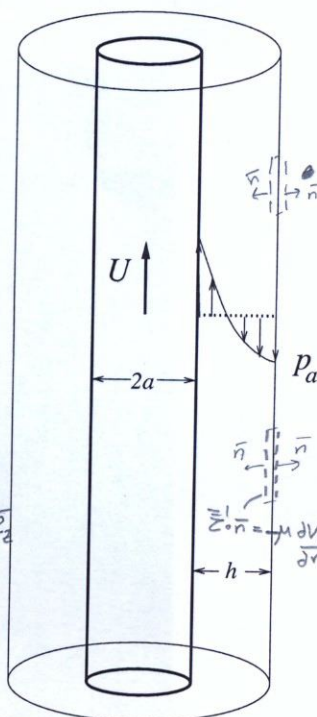
3) $Q = \int \vec{v} \cdot \vec{n} d\sigma = \int_a^{a+h} v_z 2\pi r dr =$

$= \pi U \left[(a+h)^2 - a^2 \right] + \frac{g\pi}{\nu} \left\{ \frac{(a+h)^4}{8} + \frac{h^2(2a+h)^2}{4} - \frac{a^4}{8} - \frac{(a+h)^4}{2} \ln\left(\frac{a+h}{a}\right) \right\}$

4) $0 = \phi_r + \frac{k}{r} \frac{d}{dr} \left(r \frac{\partial T}{\partial r} \right) \rightarrow T = C_1 \ln r + C_2 - \frac{\mu}{k} \left(\frac{g}{2\nu} \right)^2 \left[\frac{r^4}{16} - \frac{(a+h)^2}{2} r^2 + \frac{(a+h)^4}{2} (\ln r)^2 \right]$
 $\phi_r = \mu \left(\frac{dv_z}{dr} \right)^2 = \mu \left(\frac{g}{2\nu} \right)^2 \frac{1}{r^2} \left[r^2 - (a+h)^2 \right]^2$

$T(a) = T_c$
 $\frac{dT}{dr} \bigg|_{r=a+h} = 0 \rightarrow \text{compute!}$
 $C_1 \text{ \& } C_2$

CONDUCTIVE HEAT FLUX TO THE WALL $\rightarrow \oint \vec{q} \cdot \vec{n} d\sigma = \vec{q} \cdot \vec{n} = k \frac{\partial T}{\partial r} \bigg|_{r=a} = \frac{k C_1}{a} - f(a)$
 $\dot{Q}_w = 2\pi a \left(\frac{k C_1}{a} - f(a) \right)$



$0 = -\oint \vec{q} \cdot \vec{n} d\sigma \Rightarrow k \frac{\partial T}{\partial r} - k_a \frac{\partial T_a}{\partial r} = 0$
 $\frac{\partial T}{\partial r} = 0$

$0 = -\oint p \vec{n} d\sigma + \oint \vec{\tau} \cdot \vec{n} d\sigma$
 $\vec{\tau} \cdot \vec{n} = \mu \frac{dv_z}{dr} \vec{e}_z$

$r = a+h: p = p_a$
 $-\mu \frac{dv_z}{dr} + \mu_a \frac{dv_{za}}{dr} = 0 \Rightarrow \frac{dv_z}{dr} = 0$